John Ehrlinger, PhD

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Research Interests

Applied statistical machine learning research conducted in close collaboration with cardiovascular surgeons, cardiologists, and clinical investigators. Methodological work spans random forest and ensemble methods, clustering, deep learning, and time series analysis, with emphasis on time-to-event and longitudinal data in cardiovascular outcomes. Equally central is the communication of that work to clinical audiences, through model explainability and the visual presentation of results. Both rest on research infrastructure: clinical registry and data governance design, versioned analytic workflows, and open-source implementations that make the work reproducible at scale.

Professional Experience

Cleveland Clinic — Cardiovascular Outcomes, Registries and Research (CORR), Department of Thoracic and Cardiovascular Surgery, Heart, Vascular & Thoracic Institute

Assistant Staff, Lead Data Scientist — December 2024–Present

Lead a multidisciplinary team of data engineers, clinical data experts, and biostatisticians supporting cardiovascular outcomes research.

Methods and publication. Drive statistical methods research applied to observational clinical data, including multi-phase hazard modeling, ensemble methods, and stability-based clustering for phenotype discovery. Carry that work through to peer-reviewed publication with departmental investigators and to open-source releases on CRAN.

Research infrastructure. Lead research-side architecture and data governance for HVTR, the next-generation clinical registry platform, and direct the migration of the department's legacy SAS analytic infrastructure to reproducible, version-controlled open-source R.

Research tooling. Built the department's manuscript and proposal authoring infrastructure: a structured SharePoint editing workflow that replaced email-based document exchange and carried 30+ AATS 2026 abstract submissions, a knowledge base compiling prior publications and protocols to support proposal review, and an automated copy-editorial review tool for manuscripts under submission. Lead the group's adoption of AI for research productivity, including transcript analysis for HVTI umbrella research meetings and AI-assisted software development and documentation.

Genomics collaboration. Collaborate on the HVTI genomics and biorepository initiative, linking CORR's phenotypic registry to whole-genome sequencing and contributing to the emerging genotype–phenotype research agenda.

Cleveland Clinic Lerner College of Medicine

Assistant Professor of Surgery (Joint Appointment) — June 2025–Present

Co-advise a master's student whose machine learning research on cardiopulmonary bypass perfusion contributes directly to manuscripts in the department's active publication pipeline. Mentor cardiovascular surgery fellows in research methods, analytic best practices, and scientific communication.

Altamira Technologies

Senior Data Scientist — Technical Lead March 2023–October 2024

Led the data science team supporting USAF customers on secure networks. Built a reusable Plotly-based dashboard framework; developed predictive traffic flow models and a computer vision pipeline for vehicle volume estimation at base entry points.

Microsoft — Azure Global Commercial Industry (AGCI-AI)

Senior Data and Applied Scientist — Technical Lead July 2015–April 2023

Led customer-facing ML/AI engagements across oil & gas (Chevron), aerospace (Rolls Royce), and medical devices (Stryker). Developed Azure ML solution accelerators for predictive maintenance. Presented at Microsoft MLADS conference four times (2016–2020).

Cleveland Clinic — Quantitative Health Sciences

Assistant Staff August 2012–July 2015

Applied random survival forests to cardiovascular outcomes research. Initiated development of the ggRandomForests R package. Began migration of institutional SAS-based hazard analysis to open-source implementations.

Cleveland Clinic Lerner College of Medicine

Clinical Assistant Professor (Joint Appointment) 2015

Mentored medical students and fellows in statistical methods and research design as part of joint fellowship programs within the Cardiovascular Outcomes, Registries and Research group.

Cleveland Clinic — Thoracic and Cardiovascular Surgery

Lead Systems Analyst: Scientific Programmer August 1999–August 2012

Supported research computing infrastructure, statistical software development, and HPC operations for the department. Completed doctoral research in Statistics at Case Western Reserve University during this period.

Education

Case Western Reserve University, Ph.D. Statistics 2011

Dissertation: *Regularization: Stagewise Regression and Bagging*. Advisor: Hemant Ishwaran.

Case Western Reserve University, M.S. Mechanical / Aerospace Engineering 1994

Case Western Reserve University, B.S. Mechanical Engineering 1993

Honors & Awards

NASA Graduate Research Fellowship 1992–1994

Cleveland Clinic Innovations Award 2003

Professional Service

Moderator, CORR monthly research proposal review 2025–Present

Moderate the departmental research proposal review, and lead the proposal intake and review-process improvements adopted over the past year.

Ad hoc reviewer, *European Journal of Medical Research*

Ad hoc advisor to HVTI artificial intelligence initiatives on technology selection and clinical use-case assessment

Member, American Statistical Association and Society for Industrial and Applied Mathematics

Publications

Robinson, J., Belitsis, G., **Ehrlinger**, J., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCampli, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A. C., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., … Overman, D. M. (2026). The search for borderline hearts within biventricular repairs of complete atrioventricular septal defects. *The Journal of Thoracic and Cardiovascular Surgery*. https://doi.org/10.1016/j.jtcvs.2026.05.027

Blackstone, E. H., Chang, H. L., Rajeswaran, J., Parides, M. K., Ishwaran, H., Li, L., **Ehrlinger**, J., Gelijns, A. C., Moskowitz, A. J., Argenziano, M., DeRose, J. J., Couderc, J.-P., Balda, D., Dagenais, F., Mack, M. J., Ailawadi, G., Smith, P. K., Acker, M. A., O'Gara, P. T., … Schering, A. (2019). Biatrial maze procedure versus pulmonary vein isolation for atrial fibrillation during mitral valve surgery: New analytical approaches and end points. *The Journal of Thoracic and Cardiovascular Surgery*, *157*(1), 234–243.e9. https://doi.org/10.1016/j.jtcvs.2018.06.093

Hurst, T. E., Xanthopoulos, A., **Ehrlinger**, J., Rajeswaran, J., Pande, A., Thuita, L., Smedira, N. G., Moazami, N., Blackstone, E. H., Starling, R. C. (2019). Dynamic prediction of left ventricular assist device pump thrombosis based on lactate dehydrogenase trends. *ESC Heart Failure*, *6*(5), 1005–1014. https://doi.org/10.1002/ehf2.12473

Rajeswaran, J., Blackstone, E. H., **Ehrlinger**, J., Li, L., Ishwaran, H., Parides, M. K. (2018). Probability of atrial fibrillation after ablation: Using a parametric nonlinear temporal decomposition mixed effects model. *Statistical Methods in Medical Research*, *27*(1), 126–141.

Wojnarski, C. M., Roselli, E. E., Idrees, J. J., Zhu, Y., Carnes, T. A., Lowry, A. M., Collier, P. H., Griffin, B., **Ehrlinger**, J., Blackstone, E. H., Svensson, L. G., Lytle, B. W. (2018). Machine-learning phenotypic classification of bicuspid aortopathy. *The Journal of Thoracic and Cardiovascular Surgery*, *155*(2), 461–469. e4.

Li, L., Mao, H., Ishwaran, H., Rajeswaran, J., **Ehrlinger**, J., Blackstone, E. H. (2017). Estimating the prevalence of atrial fibrillation from a three class mixture model for repeated diagnoses. *Biometrical Journal*, *59*(2), 331–343.

Pande, A., Li, L., Rajeswaran, J., **Ehrlinger**, J., Kogalur, U. B., Blackstone, E. H., Ishwaran, H. (2017). Boosted multivariate trees for longitudinal data. *Machine Learning*, *106*(2), 277–305.

Alli, O., Rihal, C. S., Suri, R. M., Greason, K. L., Waksman, R., Minha, S., Torguson, R., Pichard, A. D., Mack, M., Svensson, L. G., Rajeswaran, J., Lowry, A. M., **Ehrlinger**, J., Tuzcu, E. M., Thourani, V. H., Makkar, R., Blackstone, E. H., Leon, M. B., Holmes, D. (2016). Learning curves for transfemoral transcatheter aortic valve replacement in the PARTNER-i trial: Technical performance. *Catheterization and Cardiovascular Interventions*, *87*(1), 154–162.

Kapadia, S., Agarwal, S., Miller, D. C., Webb, J. G., Mack, M., Ellis, S., Herrmann, H. C., Pichard, A. D., Tuzcu, E. M., Svensson, L. G., Smith, C. R., Rajeswaran, J., **Ehrlinger**, J., Kodali, S., Makkar, R., Thourani, V. H., Blackstone, E. H., Leon, M. B. (2016). Insights into timing, risk factors, and outcomes of stroke and transient ischemic attack after transcatheter aortic valve replacement in the PARTNER trial (placement of aortic transcatheter valves). *Circulation: Cardiovascular Interventions*, *9*(9), e002981.

Minha, S., Waksman, R., Satler, L. P., Torguson, R., Alli, O., Rihal, C. S., Mack, M., Svensson, L. G., Rajeswaran, J., Blackstone, E. H., Tuzcu, E. M., Thourani, V. H., Makkar, R., **Ehrlinger**, J., Lowry, A. M., Suri, R. M., Greason,

K. L., Leon, M. B., Holmes, D. R., Pichard, A. D. (2016). Learning curves for transfemoral transcatheter aortic valve replacement in PARTNER-i trial: Success and safety. *Catheterization and Cardiovascular Interventions*, *87* (1), 165–175.

Raja, S., Rice, T. W., **Ehrlinger**, J., Goldblum, J. R., Rybicki, L. A., Murthy, S. C., Adelstein, D., Videtic, G., McNamara, M. P., Blackstone, E. H. (2016). Importance of residual cancer after induction therapy for esophageal adenocarcinoma. *The Journal of Thoracic and Cardiovascular Surgery*, *152*(3), 756–761.

Suri, R. M., Alli, O., Waksman, R., Rihal, C. S., Satler, L. P., Greason, K. L., Torguson, R., Pichard, A. D., Mack, M., Svensson, L. G., Rajeswaran, J., Lowry, A. M., **Ehrlinger**, J., Mick, S. L., Tuzcu, E. M., Thourani, V. H., Makkar, R., Holmes, D., Leon, M. B., Blackstone, E. H. (2016). Learning curves for transapical transcatheter aortic valve replacement in the PARTNER-i trial: Technical performance, success, and safety. *The Journal of Thoracic and Cardiovascular Surgery*, *152*(3), 773–780. e14.

Smedira, N. G., Blackstone, E. H., **Ehrlinger**, J., Thuita, L., Pierce, C. D., Moazami, N., Starling, R. C. (2015). Current risks of HeartMate II pump thrombosis: Non-parametric analysis of interagency registry for mechanically assisted circulatory support data. *The Journal of Heart and Lung Transplantation*, *34*(12), 1527–1534.

Szeto, W. Y., Svensson, L. G., Rajeswaran, J., **Ehrlinger**, J., Smith, C. R., Mack, M., Miller, D. C., McCarthy, P. M., Bavaria, J. E., Cohn, L. H., Corso, P. J., Guyton, R. A., Thourani, V. H., Lytle, B. W., Williams, M. R., Webb, J. G., Kapadia, S., Tuzcu, E. M., Leon, M. B., Blackstone, E. H. (2015). Appropriate patient selection or healthcare rationing? Lessons from surgical aortic valve replacement in the PARTNER-i trial. *The Journal of Thoracic and Cardiovascular Surgery*, *150*(3), 557–568.

Thourani, V. H., Jensen, H. A., Babaliaros, V., Kodali, S., Rajeswaran, J., **Ehrlinger**, J., Suri, R. M., Don, C. W., Aldea, G., Blackstone, E., Williams, M. R., Makkar, R., Svensson, L., Johnson, A., McCabe, J., Dean, L. S., Johnson, M., Kapadia, S., Cohen, D. J., … Mackand, M. (2015). PARTNER: Outcomes in nonagenarians undergoing transcatheter aortic valve replacement (TAVR). *Annals of Thoracic Surgery*, *100*(3), 785–793.

Wojnarski, C. M., Roselli, E., Idrees, J., Zhu, Y., Collier, P., Griffin, B., Carnes, T., Lowry, A., **Ehrlinger**, J., Svensson, L., Blackstone, E., Lytle, B. (2015). Bicuspid valve and aneurysm phenotypes: Clinical and pathologic associations. *Journal of the American College of Cardiology*, *65*(10).

Wojnarski, C. M., Svensson, L. G., Roselli, E. E., Idrees, J., Lowry, A. M., **Ehrlinger**, J., Pettersson, G. B., Gillinov, A. M., Johnston, D. R., Soltesz, E. G., Navia, J. L., Hammer, D. F., Griffin, B., Thamilarasan, M., Kalahasti, V., Sabik, J. F., Blackstone, E. H. (2015). Aortic dissection in patients with bicuspid aortic valve– associated aneursym. *Annals of Thoracic Surgery*, *100*(5), 1666–1674.

Starling, R. C., Moazami, N., Silvestry, S. C., Ewald, G., Rogers, J. G., Milano, C. A., Rame, J. E., Acker, M. A., Blackstone, E. H., **Ehrlinger**, J., Thuita, L., Mountis, M. M., Soltesz, E. G., Lytle, B. W., Smedira, N. G. (2014). Unexpected abrupt increase in left ventricular assist device thrombosis. *New England Journal of Medicine*, *370*(1), 33–40. http://www.nejm.org/doi/full/10.1056/NEJMoa1313385

Dalton, J. E., Glance, L. G., Mascha, E. J., **Ehrlinger**, J., Chamoun, N., Sessler, D. I. (2013). Impact of present-on-admission indicators on risk-adjusted hospital mortality measurement. *Anesthesiology*, *118*(6), 1298–1306.

Ehrlinger, J., Ishwaran, H. (2012). Characterizing L_2boosting. *The Annals of Statistics*, *40*(2), 1074–1101.

In Revision / Submitted

Alshneikat, M., Alaraj, R., Awad, A. K., Ramsingh, R., **Ehrlinger**, J., Houghtaling, P. L., Koprivanac, M. J., Kapadia, S., Tong, M. Z., Soltesz, E. G., Smedira, N. G., Roselli, E. E., Wierup, P., Gillinov, A. M., Svensson, L. G., Blackstone, E. H., Pettersson, G. B., Unai, S. G., Bakaeen, F. G. (2026). Coronary artery aneurysms: Evolving trends in surgical management. *The Journal of Thoracic and Cardiovascular Surgery*.

Barron, J. O., Jain, N., Conner, A., Toth, A. J., **Ehrlinger**, J., Lee, S., Ramji, S., Sudarshan, M., Raymond, D. P., Blackstone, E. H., Murthy, S. C., Raja, S. (2026). Form over function: Development of a morphology-based achalasia classification. *The Journal of Thoracic and Cardiovascular Surgery*.

Stembal, F., Fang, M. Z., Houghtaling, P., **Ehrlinger**, J., Lieber, E., Kelava, M., Grady, P., Krzelj, K., Indorf, J.,

Somogyi, D., Troy, A., Lou, X., Kimmaliardjuk, D., Vargo, P., Malas, T., Smedira, N., Tong, M. Z., Soltesz, E., Yun, J., ... Blackstone, E. H. (2026). Is there still room for improvement of cardiopulmonary bypass? Analysis of lactate and other clinical and hemodynamic characteristics. *The Annals of Thoracic Surgery*.

In Preparation

Belitsis, G., Robinson, J., **Ehrlinger**, J., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCampli, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., ... Overman, D. (2026). *The search for borderline hearts within complete atrioventricular septal defects*.

Bhagat, R., Thuita, L., Curran, K., Blackledge, M., Raghavan, A., Zlota, A., Putnam, J., Snyder, A., Bagaber, G., Diz Ferre, J. L., Zaki, A. L., Sharma, I., Donaldson, C., **Ehrlinger**, J., Soltesz, E. G., Al-Hossan, A., Blackstone, E. H., Tong, M. Z. Y. (2026). *Who is ready for decannulation? Understanding the predictive effects of hemodynamics parameters on readiness for weaning from veno-arterial extracorporeal membrane oxygenation*.

Ehrlinger, J. (2026). Stability-based clustering for clinicians: A practical guide to SID clustering in cardiovascular research. *The Journal of Thoracic and Cardiovascular Surgery*.

Ehrlinger, J., Robinson, J., Belitsis, G., Blackstone, E. H., Li, X., Mertens, L., Fackoury, C., Williams, W. G., Karamlou, T., Jacobs, M. L., Welke, K., Jacobs, J. P., DeCampli, W., Kirklin, J. K., Pourmoghadam, K., Polimenakos, A. C., Kumar, T. K. S., Jegatheeswaran, A., Herrmann, J. L., ... Overman, D. M. (2026). *Actual vs. Optimal treatment pathway based on outcomes of complete atrioventricular septal defects using virtual twins*.

Fang, M. Z., **Ehrlinger**, J., Lieber, E., Krzelj, K., Houghtaling, P., Stembal, F., Kelava, M., Grady, P., Indorf, J., Somogyi, D., Lou, X., Kimmaliardjuk, D., Vargo, P., Malas, T., Smedira, N., Tong, M. Z.-Y., Soltesz, E., Yun, J., Zaki, A., ... Koprivanac, M. (2026). *Is cardiopulmonary bypass providing adequate perfusion? Modeling perioperative lactate levels*.

Technical Reports

Ehrlinger, J. (2016). *ggRandomForests: Exploring Random Forest Survival*. arXiv preprint. https://arxiv.org/abs/1612.08974

Ehrlinger, J. (2015). *ggRandomForests: Visually Exploring a Random Forest for Regression*. arXiv preprint. https://arxiv.org/abs/1501.07196

Conference Presentations

Ehrlinger, J. (2026). *Care and Feeding of Your Biostats Team: Scaling Best Practices in a Large Hybrid SAS/R Team*. Invited talk, R/Medicine Conference. https://ehrlinger.github.io/CareFeedingBiostats/

Zamanian, F., **Ehrlinger**, J., Zhan, C. (2020). *How to interpret model prediction using interpretability tools on Azure Machine Learning*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

Ehrlinger, J., Solanki, P. S. (2019). *Distributed Strategies for Training Deep Learning Models at Scale*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

Ehrlinger, J., Shah, P. (2018). *Automated Machine Learning: A Case Study in Efficiently Expanding Data Science Experimentation for Stryker Surgical IoT Devices*. Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

Ehrlinger, J. (2016). *Are we counting correctly? Model evaluation for time series classification.* Conference session, Microsoft Machine Learning and Data Science (MLADS), Redmond, WA.

Ehrlinger, J. (2014). *Visually Exploring Random Forests with ggRandomForests*. Conference session, useR! 2014, UCLA, Los Angeles, CA.

Software

R Packages (CRAN)

ggRandomForests — Visual exploration of random forest models — graphical analysis of survival, regression, and classification forests. v3.5.2 on CRAN. https://CRAN.R-project.org/package=ggRandomForests

TemporalHazard — R port of the C computational core underlying the Cleveland Clinic Hazard SAS module. v1.1.0 on CRAN. https://CRAN.R-project.org/package=TemporalHazard

HVTI R Package Family

Eleven member packages — the nine below, plus *ggRandomForests* and *TemporalHazard* above — resolved from public GitHub repositories and installed, updated and version-checked as a unit by *hvtiR*, a one-command installer and environment diagnostic. https://github.com/ehrlinger/hvtiR

hvtiBoostmtree — Boosted multivariate trees for longitudinal data; an extended fork of boostmtree. https://github.com/ehrlinger/hvtiBoostmtree

hvtiPlotR — HVTI-standard publication graphics for reproducible clinical research figures. https://github.com/ehrlinger/hvtiPlotR

hvtiRbootstrap — Bootstrap model building — fit across many replicates and report how often each variable survives selection; an R port of the bootreg, SUMBOOT and cluster SAS macros; in active development. https://github.com/ehrlinger/hvtiRbootstrap

hvtiRdatabuild — Analysis-ready clinical datasets for HVTI CORR studies, verified against the legacy SAS datasets they replace; in active development. https://github.com/ehrlinger/hvtiRdatabuild

hvtiRlifetables — Age-, sex- and race-matched US reference survival; replaces the usmatchd SAS macro by evaluating a stored three-phase parametric hazard fit rather than interpolating a life table; in active development. https://github.com/ehrlinger/hvtiRlifetables

hvtiRpropensity — Propensity score estimation, matching and IPTW with standardized balance diagnostics, for cardiac surgery comparative-effectiveness research; in active development. https://github.com/ehrlinger/hvtiRpropensity

hvtiRtables — Manuscript-compliant Word tables from gtsummary objects, following HVTI CORR table construction standards, with a JTCVS submission mode. https://github.com/ehrlinger/hvtiRtables

hvtiRtemplates — Versioned analysis job templates and the analysis-prefix taxonomy the CORR group organizes jobs by, so a study binds to a versioned template rather than to a copy. https://github.com/ehrlinger/hvtiRtemplates

hvtiRutilities — Utility functions supporting reproducible HVTI research workflows. https://github.com/ehrlinger/hvtiRutilities

Open-Source Documentation

HVTI Recipes — Catalog of publication-ready figures, tables, and datasets for clinical outcomes research — Kaplan-Meier, propensity balance, CONSORT, random-forest visualizations — each paired with reproducible code. Quarto book, CC BY 4.0. https://ehrlinger.github.io/hvtiGraphics/

SAS/C Software

hazard — SAS and C implementation of multi-phase hazard analysis for time-to-event decomposition. (Maintainer) https://github.com/ehrlinger/hazard